

Lecture 27: Alternative Water Reactors: The BWR

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 The Boiling Water Reactor (BWR)

The BWR (approx. 15% - 20% of the global fleet) utilizes a **Direct Cycle**: light water boils directly in the core, and that same steam drives the turbine.

1.1 Circulation and Steam Separation

Contrary to the simple "pot on a stove" model, the water in a BWR is vigorously recirculated.

1. **The Recirculation Ratio:** Only a small fraction (approx. 14% by mass) of the water entering the core actually turns to steam in a single pass. The rest remains liquid.
2. **Steam Separation:** The two-phase mixture (liquid + vapor) exits the core and passes through **Steam Separators** (cyclones) and **Dryers** (chevron vanes) at the top of the vessel to remove moisture droplets.
 - *Steam Path:* Dry steam exits the vessel and goes directly to the High Pressure Turbine.
 - *Liquid Path:* The removed water flows down the "downcomer" (annulus between core shroud and vessel wall), mixes with fresh feedwater, and is pumped back up through the core.

1.2 Standard Recirculation: The Jet Pump System

Most operating BWRs (e.g., BWR/4, BWR/5) use **Forced Circulation** driven by **Jet Pumps** located inside the vessel annulus.

- **The Mechanism:** This is very similar to a residential "deep well" ejector pump.
- **Operation:** External recirculation pumps pull a small amount of water out of the vessel (the "driving flow") and shoot it back in at high velocity through a nozzle. This high-velocity jet entrains the surrounding water (the "suction flow") and drags it into the core.
- **Control Strategy:** Reactor power is controlled by changing the speed of the external pumps.

Faster Flow $\rightarrow$ Sweeps bubbles out $\rightarrow$ Void fraction $\downarrow \rightarrow$ Moderation $\uparrow \rightarrow$ **Power Rises**

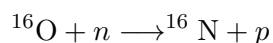
1.3 Reactor Core Isolation Cooling (RCIC)

The RCIC system is a critical, dedicated emergency cooling system designed to maintain water inventory in the reactor core if it becomes isolated from its primary heat sink (the main steam lines to the turbine and condenser).

- **Steam-Driven Motive Force:** The most vital engineering feature of the RCIC is that its main pump does not rely on the plant's massive AC electrical grid or emergency diesel generators. Instead, it taps into a main steam line and uses a small bleed of the high-pressure steam (generated by the reactor's own decay heat) to spin a dedicated turbine.
- **Water Injection:** This steam turbine is mechanically coupled directly to a water pump. As the turbine spins, the pump draws cold reserve water (typically from the Condensate Storage Tank or the Suppression Pool in the torus) and injects it back into the reactor vessel at high pressure to replace boiled-off inventory.
- **Exhaust:** The steam used to spin the RCIC turbine is then piped down into the Suppression Pool (the wetwell) where it bubbles up and condenses.
- **The DC Power Vulnerability:** While the pump itself is mechanically driven by steam, the motorized control valves and electronic instrumentation required to operate the RCIC system rely entirely on DC battery power. If DC power is lost (as happened in the flooded basements of Fukushima), the valves cannot be actuated remotely, effectively blinding and disabling this otherwise robust mechanical system unless operators can jury-rig external batteries.

1.4 Operational Challenges

1. **Radioactive Turbine (N-16):** Because the primary coolant goes to the turbine, the steam contains Nitrogen-16, produced by fast neutron capture in oxygen:



^{16}N emits very high-energy gammas (6 MeV and 7 MeV). The turbine hall is heavily shielded during operation, but because $T_{1/2} = 7.13$ seconds, radiation levels drop to safe limits within minutes of shutdown.

2. **No Chemical Shim:** We cannot dissolve Boric Acid in the coolant for reactivity control both because it would plate out on the fuel and turbine blades when the water boils, and concentration variations would make the effect on reactivity highly variable. Reactivity is managed solely by flow control and control rods.
3. **Bottom-Entry Control Rods:** The top of the reactor vessel is crowded with steam dryers. Therefore, control blades (cruciform shape) enter from the bottom. Gravity cannot scram the reactor; rods must be driven in hydraulically using high-pressure accumulators.

2 BWR Containment Strategy and Fukushima

Unlike the massive "Large Dry" domes of PWRs, BWRs rely on a thermodynamic trick called **Pressure Suppression** to manage a LOCA.

2.1 The Mark I Containment (Pressure Suppression)

The original GE design (used at Fukushima Daiichi Units 1-5) is the Mark I, famous for its "light-bulb and torus" geometry.

- **Drywell (The Lightbulb):** A steel pressure vessel immediately surrounding the reactor. If a primary pipe breaks, expanding steam fills this confined space.
- **Wetwell (The Torus):** A massive donut-shaped tube running around the base of the reactor, half-filled with millions of gallons of cold water (the Suppression Pool).
- **The Mechanism:** Massive vent pipes connect the Drywell to the Wetwell. During a LOCA, expanding steam is forced down the vents and bubbled through the cold water of the Suppression Pool, instantly condensing it and drastically reducing peak accident pressure.

2.2 The Fukushima Daiichi Accident (2011)

The accident was fundamentally a prolonged Station Blackout (SBO) leading to a complete loss of the ultimate heat sink.

1. **The Earthquake and SCRAM:** On March 11, a magnitude 9.0 earthquake struck. The reactors successfully SCRAMmed. Off-site power was lost, but emergency diesels started automatically.
2. **The Tsunami and SBO:** 45 minutes later, a 14-meter tsunami overtopped the plant's 5.7-meter seawall. The plant operator, TEPCO, actually knew from historical data and internal studies that a massive tsunami was possible, but had failed to act, leaving their seawall nearly 10 meters too short. The water violently flooded the basements where the diesel generators and DC batteries were housed, causing a total SBO.
3. **Loss of Cooling (Unit 1 vs. Units 2/3):** Unit 1's passive Isolation Condenser was valved shut prior to the tsunami to prevent thermal shock, and could not be reopened without DC power, leading to a rapid meltdown. In Units 2 and 3, operators heroically kept the steam-driven RCIC systems running for days by scavenging car batteries, delaying core damage.
4. **Core Melt, Vessel Breach, and Hydrogen Generation:** Once cladding temperatures reached $\approx 1200^\circ\text{C}$, the highly exothermic Zircaloy-steam reaction initiated, generating massive amounts of explosive hydrogen gas. The uncooled fuel eventually melted completely, forming a molten mass called *corium*. This corium melted through the bottom of the steel Reactor Pressure Vessels (RPVs) in Units 1, 2, and 3, dropping onto the concrete floor of the primary containment (Drywell) and breaching the primary boundaries.
5. **Containment Venting and Explosions:** To prevent the primary containment from rupturing entirely due to overpressure, operators vented the gases. The vented hydrogen backed up into the secondary containment (reactor buildings) and detonated, blowing the roofs off Units 1, 3, and 4. Unit 4 was completely defueled and cold at the time, but was destroyed because it shared a ventilation exhaust stack with Unit 3; hydrogen flowed backward into Unit 4's building and ignited.
6. **Radioactive Release and the TMI Contrast:** The combination of vessel breaches by the corium, containment seal failures, and hydrogen explosions led to a significant release of radioactive fission products. This sharply contrasts with the 1979 Three Mile Island (TMI)

accident. TMI also suffered a core meltdown, but released virtually zero radioactivity because its massive "Large Dry" PWR containment dome easily absorbed the hydrogen burn and pressure spikes. Building a TMI-style dome large enough to house a BWR and its towering steam separation equipment was deemed economically unviable in the 1960s. Instead, GE relied on the smaller, cheaper Mark I pressure suppression system, a cost-saving compromise that ultimately failed to contain a severe, multi-unit accident.

2.3 Health Impacts and Casualties: The Three Major Accidents

A common misconception is that the dramatic hydrogen explosions at Fukushima resulted in mass casualties similar to the 1986 Chernobyl disaster. It is crucial to contrast the physical realities and health impacts of the three major nuclear accidents to understand the importance of containment:

- **Fukushima (2011) - Bypassed Containment:** Zero plant workers died in the hydrogen explosions. Two operators in Unit 4 were killed by the initial tsunami wave that flooded the basement prior to any meltdowns. While about 16 workers were injured by debris during the blasts, no acute radiation deaths occurred. The vast majority of the disaster-related deaths (estimated to be over 2,000) were caused by the chaotic mass evacuation, where elderly and critically ill hospital patients died from severe stress and lack of medical care.
- **Chernobyl (1986) - No Containment:** Chernobyl suffered a criticality and steam explosion that completely destroyed the reactor and exposed the burning graphite core to the open atmosphere for days, releasing roughly ten times more radioactive material than Fukushima. As a result, 28 plant workers and first responders died within weeks from Acute Radiation Syndrome (ARS), in addition to two killed in the initial blast. While the highly organized Soviet evacuation prevented immediate transit deaths, the permanent displacement of over 350,000 people likely resulted in a massive long-term indirect death toll driven by depression, alcoholism, and severe stress-related illnesses.
- **Three Mile Island (1979) - Successful Containment:** The partial meltdown of the TMI Unit 2 PWR provides the ultimate contrast. Despite a severe core melt and a hydrogen burn inside the reactor building, the massive "Large Dry" containment dome remained entirely intact. The radioactive release to the environment was negligible (primarily trace amounts of noble gases). There were zero deaths, zero acute injuries, and decades of epidemiological studies have found no statistically significant increase in cancer rates in the surrounding population. TMI proved that a robust containment structure can successfully protect public health even during a catastrophic core failure.

3 Post-Fukushima Regulatory Response

As is often the case in engineering, nuclear regulations are fundamentally reactive. After a major accident, the industry thoroughly analyzes the specific failure modes and mandates systemic fixes to ensure that exact sequence of events cannot happen again, and that hopefully the fixes will be broad enough that related disasters will be avoided as well.

3.1 The FLEX Strategy (Diverse and Flexible Coping Strategies)

To combat a prolonged Station Blackout (SBO) and the loss of the ultimate heat sink, the U.S. NRC mandated a tiered, defense-in-depth approach known as FLEX.

- **Phase 1 (Installed Equipment):** Rely on permanently installed plant equipment (like DC batteries and the steam-driven RCIC pumps) to maintain core cooling for the first several hours.
- **Phase 2 (On-Site Portable Equipment):** Transition to portable, diesel-driven pumps and generators. Plants are now required to store this equipment in hardened, decentralized concrete bunkers on-site, with pre-installed standardized hookups (ensuring workers never have to jury-rig car batteries again).
- **Phase 3 (Off-Site National Response):** If the site is utterly destroyed, the industry established two National SAFER Response Centers (in Phoenix, AZ, and Memphis, TN). These centers maintain massive stockpiles of heavy generators, pumps, and water purification units that can be airlifted to any U.S. plant within 24 hours.

3.2 Severe Accident Capable Hardened Vents (HCVS)

Because the hydrogen explosions at Fukushima Units 1, 3, and 4 were caused by explosive gases backing up into the secondary containment (reactor buildings), the NRC issued Order EA-13-109. All Mark I and Mark II BWRs were required to install hardened containment vents. These robust pipes allow operators to vent pressure and explosive gases directly to the outside atmosphere, safely bypassing the reactor building entirely, even during a total blackout with active core damage.

3.3 Spent Fuel Pool Instrumentation

During the Fukushima crisis, operators lost all instrumentation for the Spent Fuel Pools (SFPs) and did not know if the spent fuel was boiling dry and catching fire (it wasn't, but the uncertainty caused global panic and delayed response efforts). The NRC subsequently mandated independent, battery-backed water level instrumentation for all SFPs to ensure operators are never again flying blind during a severe accident.

4 Economics, Time, and the Derailed "Nuclear Renaissance"

The early 2000s saw a renewed push for nuclear power in the U.S., commonly dubbed the "Nuclear Renaissance." However, this momentum was shattered by a combination of the Fukushima disaster and brutal domestic economic realities.

4.1 The Vulnerability of Long Lead Times

Nuclear power plants are massive capital megaprojects. The timeline from initial site planning and NRC licensing, to securing multi-billion dollar financing, pouring concrete, and finally commissioning the reactor often spans 10 to 15 years (or more).

- **Market Shifts:** This massive lead time makes nuclear construction extremely vulnerable to shifting macroeconomic conditions. A plant design that makes perfect economic sense when proposed in 2005 might be financially ruinous by the time it is ready to generate power in 2018.

4.2 The Shale Gas Boom and Cancellations

During the exact decade that the "Nuclear Renaissance" plants were trying to get off the ground, the U.S. experienced the shale gas "fracking" boom.

- **The Economic Squeeze:** Fracking drove domestic natural gas prices to historic lows. Simultaneously, the post-Fukushima environment resulted in more stringent safety reviews, stretching out timelines and increasing capital costs.
- **The Result:** This one-two punch of cheap fossil fuel competition and increased regulatory burden made the massive upfront capital requirements of new BWRs unviable. Several planned U.S. BWR expansions were officially canceled in the 2010s, including two new Advanced BWR (ABWR) units at the South Texas Project and the planned Fermi Unit 3 (an ESBWR) in Michigan.

4.3 The Asian Market: Public Trust and Political Cancellations

While the U.S. cancellations were primarily driven by the economics of the natural gas boom, the derailment of the nuclear renaissance in Asia was a direct result of the post-Fukushima political climate and public opposition.

- **Japan's Complete Shutdown:** Following the accident, public trust was entirely shattered. Japan shut down its entire fleet of 54 operating reactors for safety reviews. While a fraction have slowly restarted under a strict new regulatory authority, many older units were permanently decommissioned, and planned domestic expansions were completely scrapped.
- **Taiwan and the Lungmen Project:** The political shockwave was acutely felt in neighboring Taiwan. Massive public protests led the government to halt the Lungmen Nuclear Power Plant. This plant consisted of two nearly completed Generation III ABWRs. Billions of dollars in sunk capital were effectively abandoned, and the reactors were mothballed due to political pressure rather than technological or strictly economic failure.
- **The Regional Pause:** Other major nuclear players, such as South Korea and China, instituted immediate pauses on new reactor approvals to conduct sweeping safety reviews and recalibrate their energy policies, significantly stalling the region's momentum for nuclear growth for several years.

5 Advanced Generation III/III+ BWRs

To address the vulnerabilities exposed by older designs, the industry developed advanced reactors focusing on eliminating large LOCA pathways and surviving total blackouts.

5.1 The Advanced Boiling Water Reactor (ABWR)

The ABWR is an evolutionary Generation III design and the only advanced BWR to see commercial operation (four units built in Japan prior to 2011).

- **Internal Recirculation (RIPs):** The defining feature of the ABWR is the elimination of the large external recirculation loops and jet pumps. Instead, it uses **Reactor Internal Pumps (RIPs)** mounted directly through the bottom of the reactor vessel.

- **LOCA Mitigation:** By keeping all primary coolant piping essentially within the vessel itself, the possibility of a large-break LOCA below the core elevation is physically eliminated, massively increasing the safety margin.
- **Commercial Fate:** Although all of the ABWR projects were shut down or abandoned after Fukushima, the reactor design is coming back to life again. While its global expansion was severely stunted—ambitious export projects in the U.S. (South Texas Project), the UK (Wylfa Newydd), and Taiwan (Lungmen) were all ultimately canceled due to shifting economics and political opposition—the design survived in Japan. Following a grueling, 15-year regulatory battle and massive safety upgrades, TEPCO successfully restarted the Kashiwazaki-Kariwa Unit 6 ABWR in early 2026, marking the utility’s first operational reactor since the Fukushima disaster.

5.2 The Economic Simplified Boiling Water Reactor (ESBWR)

The ESBWR is a Generation III+ design intended to be entirely ”walk-away safe” and immune to a Fukushima-style blackout.

- **Natural Convection:** The ESBWR eliminates recirculation pumps entirely. Flow is driven purely by natural circulation. As water boils, the steam bubbles decrease fluid density (increasing the void fraction). The dense, cold liquid in the downcomer naturally pushes the lighter two-phase mixture up through a tall chimney added above the core.
- **Void Coefficient Control:** Because it relies on bubbles to move fluid, the reactor exhibits a very strong negative void coefficient, naturally balancing its own power output. Power is controlled by adjusting feedwater temperature and via Fine Motion Control Rod Drives (FMCRD).
- **Passive Safety (The 72-Hour Rule):** Utilizing a Gravity-Driven Cooling System (GDCS) and passive Isolation Condensers, the ESBWR is licensed to safely cool itself during a total SBO for a minimum of **72 hours** without any AC power or operator intervention.
- **Commercial Fate:** Despite receiving full NRC design certification in 2014, a full-scale commercial ESBWR was never constructed due to the economic factors mentioned above.

5.3 The BWRX-300 Small Modular Reactor (SMR)

While the massive 1500+ MWe ESBWR was never built, its passive technology did not die.

- GE-Hitachi scaled down the ESBWR’s natural circulation and passive safety concepts to create the **BWRX-300**, a 300 MWe Small Modular Reactor.
- By drastically reducing the physical size, capital cost, and construction time, the BWRX-300 aims to overcome the financial hurdles of the previous decade. It is currently leading the global SMR race, with the first unit actively under construction at the Darlington site in Ontario, Canada.

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